The Wings and the Belly

SVI as a moment-controlled smile: Lee's bound, the tails it certifies, and the interior it does not

Vol-Fitter Technical Notes, No. 02 — lecture edition

Vol-Fitter

July 24, 2026

Abstract

A volatility smile answers two questions at once. Its *wings* report the tails of the risk-neutral distribution — how much mass sits far out of the money — and Lee's moment formula caps how fast they may grow. Its *belly*, the curvature near the money, reports the shape of the centre. This note reads raw SVI through that division. SVI draws one expiry's total implied variance as a tilted hyperbola: strictly convex, linear in both wings, and unable to turn twice. We show that its two asymptotic wing slopes $\beta_L = b(1-\rho)$ and $\beta_R = b(1+\rho)$ are exactly the object Lee's bound constrains, and that the bound lands inside Durrleman's density factor as the elementary tail limit $g(\pm\infty) = (4-\beta^2)/16$: the wing slope sets the *sign* of the implied density in the tail, and staying *strictly* under Lee's 2 is precisely the condition that the tail density come out eventually positive — the boundary $\beta = 2$ itself is a trap (proposition 2). Every cheap no-arbitrage guarantee raw SVI can give therefore lives in the wings. The belly is where it cannot: a Lee-clean, positive, strictly convex slice (Axel Vogt's classical counterexample — minimum variance 0.0116, larger Lee coefficient 0.174, both tails of g at $\approx 0.25 > 0$) still drives g to -0.033 in its interior, so its Black density is negative near $k \approx 0.88$. The jump-wing (SVI-JW) coordinates are the desk's way of naming this split: two *tail* handles p, c and three *belly* handles $v, \psi, \widetilde{v}$. We prove the exact regular domain on which the five handles invert to a raw slice, and show that where they fail to identify it — the $\psi = 0$ stratum — what they lose is precisely the belly curvature: infinitely many raw widths share the same tails and level. We then fit the slice the way the split suggests: an unconstrained reparametrization that fences the tails (the Lee and minimum rows) while the data term fits the belly, with a closed-form Levenberg–Marquardt Jacobian that recovers a production benchmark through a JW→raw→fit→JW round trip to 5.6×10^{-13} vol bp and runs 2.40× faster than finite differences. Product status up front: production fits and stores *raw* SVI and displays model-agnostic ATM handles; five-handle JW entry, bumping and export are analytical, not shipped.

Contents

1 Two questions a smile answers

Stand at one expiry and look at the implied-volatility smile as a curve over log-moneyness. It is doing two jobs. Far from the money it is reporting the *tails* of the risk-neutral law — the price the market puts on rare moves — and near the money it is reporting the *belly*, the local shape around the forward. These two jobs are governed by different mathematics. The tails obey a hard theorem: Lee's moment formula ties the growth rate of implied variance in the wings to the number of finite moments of the underlying, and caps that growth. The belly obeys no such cheap law; it is where a smile can look perfectly reasonable and still be arbitrageable.

The organizing claim of this note is that raw SVI wears this division on its sleeve, and that every guarantee the Vol-Fitter cheaply enforces on an SVI slice is a *wing* guarantee. Once that is clear, three otherwise separate facts become one story: why the coded no-arbitrage screens are tail conditions, why the classical counterexample that defeats them fails in the *belly*, and why the trader coordinate system (SVI-JW) is built to separate the two. We develop the geometry first, read the tails through Lee's bound, watch the belly escape, and only then change coordinates and fit.

Conventions and clocks. Throughout, $k = \log(K/F_T)$ is log-moneyness against the expiry-T forward F_T, and

$$w(k) = (\text{Black implied total variance at } k)$$

is the object SVI models. The normalized Black price depends only on k and w, so w is recovered from a quoted price without first choosing an annualization clock. Vol-Fitter keeps two clocks apart:

$$I_t(k) = \sqrt{\frac{w(k)}{t}}, \qquad I_\tau(k) = \sqrt{\frac{w(k)}{\tau}}, \tag{1}$$

where $t = t_T$ is the plain calendar year fraction and $\tau = \tau_T$ is the *active variance time* — calendar time dilated by scheduled events (Notes 00 and 11), equal to t when the event clock is off. Production passes τ to the calibrator, so unless qualified I means the working quantity I_τ; changing the clock rescales the annualized number, never w. A *volatility basis point* (vol bp) is 10^{-4} in volatility. We write the SVI width as s so that σ is free for volatility, and reserve one differentiation convention: a prime denotes the derivative of a one-variable function (w', w'', I'), a subscripted or explicit ∂ denotes a partial ($\partial w/\partial\rho$).

Symbol	Meaning	First used
k	log-moneyness $\log(K/F_T)$	(1)
$w(k)$	total implied variance (price-derived)	(2)
$t,\ \tau$	calendar year fraction; active variance time	(1)
$I(k)$	working implied volatility $\sqrt{w/\tau}$	(1)
a, b, ρ, m, s	raw SVI parameters (width s)	(2)
β_L, β_R	asymptotic wing slopes $b(1 \mp \rho)$ of w	(6)
$k_\star,\ w_\star$	minimizing k; minimum total variance	(5)
$g(k)$	Durrleman density factor	(7)
$v, \psi, p, c, \widetilde{v}$	JW handles: level, ATM slope, wings, floor	(11)
χ	normalized displacement $m/\sqrt{m^2 + s^2}$	(15)
D	inverse denominator (an angle gap)	(16)
θ	unconstrained calibration vector in $\mathbb{R}^5$	(21)

Table 1: Notation ledger. Nothing is reused with a second meaning; the width is s, never σ, and there is one time symbol per clock.

Raw SVI (Gatheral) postulates that the entire slice is one hyperbola:

$$\boxed{w(k) = a + b\Big\{\rho(k - m) + \sqrt{(k - m)^2 + s^2}\Big\}.} \tag{2}$$

Five parameters, and — as we are about to see — exactly two of them, bundled as β_L, β_R, are what Lee's bound touches. The rest describe the belly.

Invariant

1. Every finite $\theta \in \mathbb{R}^5$ maps to a smooth raw hyperbola with $b > 0$, $|\rho| < 1$, $s > 0$ (section 6); positivity and no-arbitrage need more.
2. The two coded screens are cheap global fences: the vertex floor $\widetilde{v} \geq 0$ (a belly condition) and the *strictly buffered* wing cap $\max(\beta_L, \beta_R) \leq \beta_{\max} = 1.95 < 2$ (the boundary $\beta = 2$ itself admits negative tail density, proposition 2). Both are finite-weight soft rows, exactly zero on a clean slice, and they bound but do not certify non-negativity of the Black density in the belly.
3. The raw↔JW change of coordinates is exact on the regular domain of theorem 1; the singular stratum it omits is a loss of *belly* information (remark 1).
4. SVI is one display overlay among several: band, calendar, var-swap, prior and extrapolation blocks share the series' product targets through model-specific residuals, and each is byte-identical to its own absence.

2 One hyperbola with straight wings

Everything below rests on three closed-form facts about (2). Write

$$x = k - m, \qquad r = \sqrt{x^2 + s^2}, \qquad q = \sqrt{1 - \rho^2}. \tag{3}$$

Proposition 1 (Geometry of one slice)**.** *Under $b > 0$, $|\rho| < 1$, $s > 0$, the slice w is strictly convex with a single minimum:*

$$w'(k) = b\Big(\rho + \frac{x}{r}\Big), \qquad\qquad w''(k) = \frac{b\, s^2}{r^3} > 0, \tag{4}$$

$$k_\star = m - \frac{s\rho}{q}, \qquad\qquad w_\star = a + b\, s\, q. \tag{5}$$

Its far field is two straight rays,

$$w(k) = \beta_L\, |k| + O(1)\ (k \to -\infty), \quad w(k) = \beta_R\, k + O(1)\ (k \to +\infty), \quad \beta_L = b(1-\rho),\ \beta_R = b(1+\rho). \tag{6}$$

Proof. Differentiating (2) gives (4); since $b, s > 0$, $w'' > 0$ for every finite k, so a stationary point is the unique global minimum. Solving $w' = 0$ gives $x = -s\rho/q$, hence $r = s/q$ and (5). In either wing $r = |x| + o(1)$, which yields (6). □

Two consequences are worth stating plainly, because they drive the rest of the note. First, $w'' > 0$ *everywhere*: raw SVI is one globally convex curve with one belly and one turn of the slope, from $-\beta_L$ up to $+\beta_R$ (fig. 1A). This is the model's rigidity, and section 8 returns to it. Second — the point of this section — the tails collapse to *two numbers*. As $|k|$ grows, $w(k)/|k|$ converges to β_L on the put side and β_R on the call side (fig. 1B); no other feature of the slice survives to infinity. Whatever the market's tails are worth, SVI records them in β_L and β_R alone.

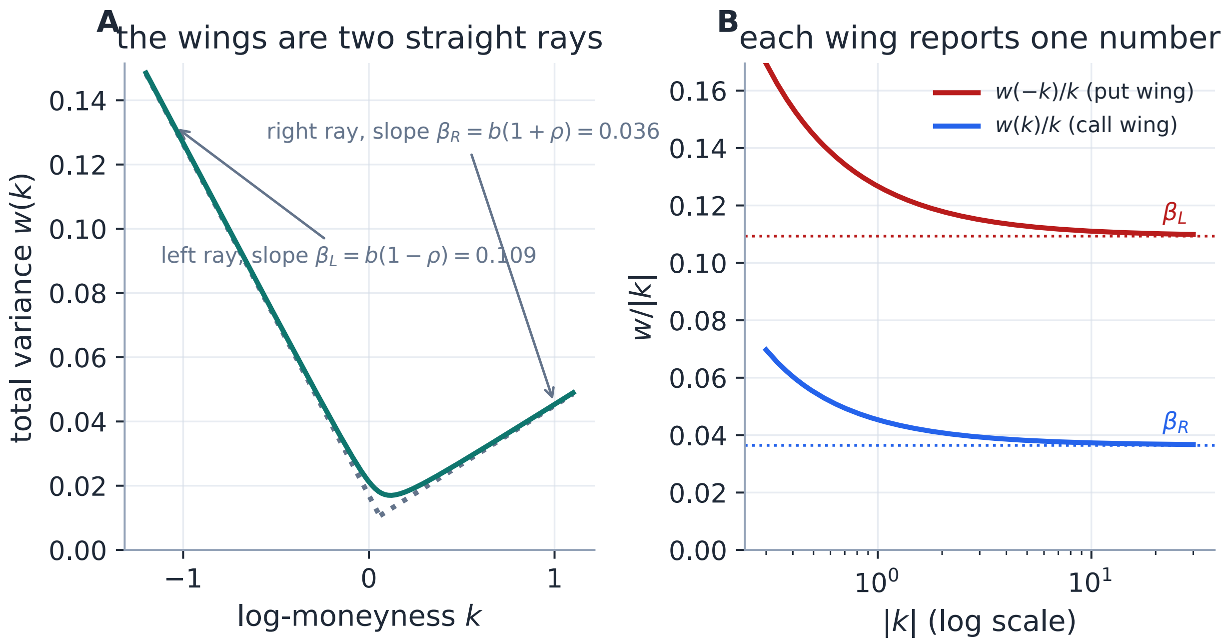

Figure 1: The wings are a two-number reading of the tails. Panel A: the total variance approaches two straight asymptotes, steeper on the put side for the equity skew $\rho < 0$. Panel B: $w(k)/|k|$ collapses onto the constants β_L and β_R as $|k| \to \infty$ — each wing reports exactly one slope, and those two slopes are all that Lee's bound of section 3 can see.

Heuristic

Read (2) as a rounded hinge. Far out, the hinge has forgotten its rounding and shows two straight rays of slope β_L (left) and β_R (right). Near the centre, s decides how sharply one ray turns into the other, m sets where the rounding sits, and a raises the whole curve. The tilt ρ divides a fixed total steepness $\beta_L + \beta_R = 2b$ between the two sides. The wings are tails; the region where s matters is the belly.

3 Lee's bound is a statement about the wings

Why should a wing slope be capped at all? Because it is a moment barometer. Lee's moment formula [3] says that if the underlying has $\bar{p}$ finite moments beyond the first, the right-wing implied variance grows no faster than $\beta_R k$ with a slope controlled by $\bar{p}$, and symmetrically on the left; in the limiting case of a distribution on the edge of losing all moments, the slope reaches 2. Heavier tails would demand a steeper wing, and a wing steeper than 2 is not the tail of *any* probability distribution. For a model whose wings are exactly linear, this is not an asymptotic nicety — it is a sharp, checkable inequality on two of its parameters.

The cleanest way to see the bound bite is to follow it inside the object that actually tests for arbitrage. A positive, C^2 total-variance slice with the right wing behaviour is free of butterfly arbitrage iff the Black density it implies is non-negative, and Durrleman [4] writes that density, up to a strictly positive prefactor, as

$$g(k) = \Big(1 - \frac{k\,w'(k)}{2w(k)}\Big)^2 - \frac{w'(k)^2}{4}\Big(\frac{1}{w(k)} + \frac{1}{4}\Big) + \frac{w''(k)}{2} \;\geq\; 0 \quad \forall k. \tag{7}$$

The connection to the wings is a one-line limit.

Proposition 2 (The wing slope sets the sign of the tail density)**.** *For a structurally valid positive raw slice,*

$$\lim_{k\to-\infty} g(k) = \frac{4-\beta_L^2}{16}, \qquad \lim_{k\to+\infty} g(k) = \frac{4-\beta_R^2}{16}. \tag{8}$$

Hence for $\beta < 2$ the tail of g is eventually positive, for $\beta > 2$ eventually negative — and at $\beta = 2$ *the limit vanishes and decides nothing: the sign falls to the next order. On a right wing at the boundary, $w(k) = 2k + \alpha + O(k^{-1})$ (for raw SVI with $\beta_R = 2$, $\alpha = a - 2m$),*

$$g(k) = \frac{\alpha - 2}{4k} + O(k^{-2}), \tag{9}$$

so the boundary slice carries negative *tail density whenever $\alpha < 2$ (mirror statement on the left with $\alpha' = a + 2m$). The call-price boundary* $\lim_{k\to+\infty} d_+(k) = -\infty$ *(equivalently, the normalized call vanishes at infinite strike) is strict as well: it holds exactly when $\beta_R < 2$.*

Proof. In either wing $w \sim \beta|k|$, $|w'| \to \beta$, and $w'' \to 0$ by (4). The first bracket of (7) tends to $1 - \frac{1}{2} = \frac{1}{2}$, so its square tends to $\frac{1}{4}$; the middle term tends to $\beta^2/16$ because $1/w \to 0$; the last term vanishes. This gives (8). For the boundary, with $d_+(k) = -k/\sqrt{w} + \frac{1}{2}\sqrt{w}$ and $w \sim \beta_R k$ on the right,

$$d_+(k) = \sqrt{k}\Big(-\tfrac{1}{\sqrt{\beta_R}} + \tfrac{\sqrt{\beta_R}}{2}\Big) + o(\sqrt{k}),$$

whose leading coefficient is negative exactly for $\beta_R < 2$; at $\beta_R = 2$ the two terms cancel and $d_+ \to 0$. For the boundary order, write the right wing at $\beta_R = 2$ as $w(k) = 2k + \alpha + O(k^{-1})$. Then $kw'/(2w) = \frac{1}{2} - \alpha/(4k) + O(k^{-2})$, so the first bracket of (7) squares to $\frac{1}{4} + \alpha/(4k)$; the middle term is $\frac{1}{4} + 1/(2k)$ up to $O(k^{-2})$ because $w'^2 \to 4$ and $1/w = 1/(2k) + O(k^{-2})$; and w'' contributes $O(k^{-3})$. Subtracting gives (9). □

Figure 2 makes the mechanism visible. The map $\beta \mapsto (4-\beta^2)/16$ is a downward parabola crossing zero at Lee's cap $\beta = 2$: below the cap the tail of g is a positive constant, above it a negative one. Carried into a concrete slice, a wing with $\beta = 1.6$ settles onto a positive plateau far out, while a wing with $\beta = 2.3$ settles onto a negative one — a butterfly arbitrage that lives arbitrarily deep in the tail.

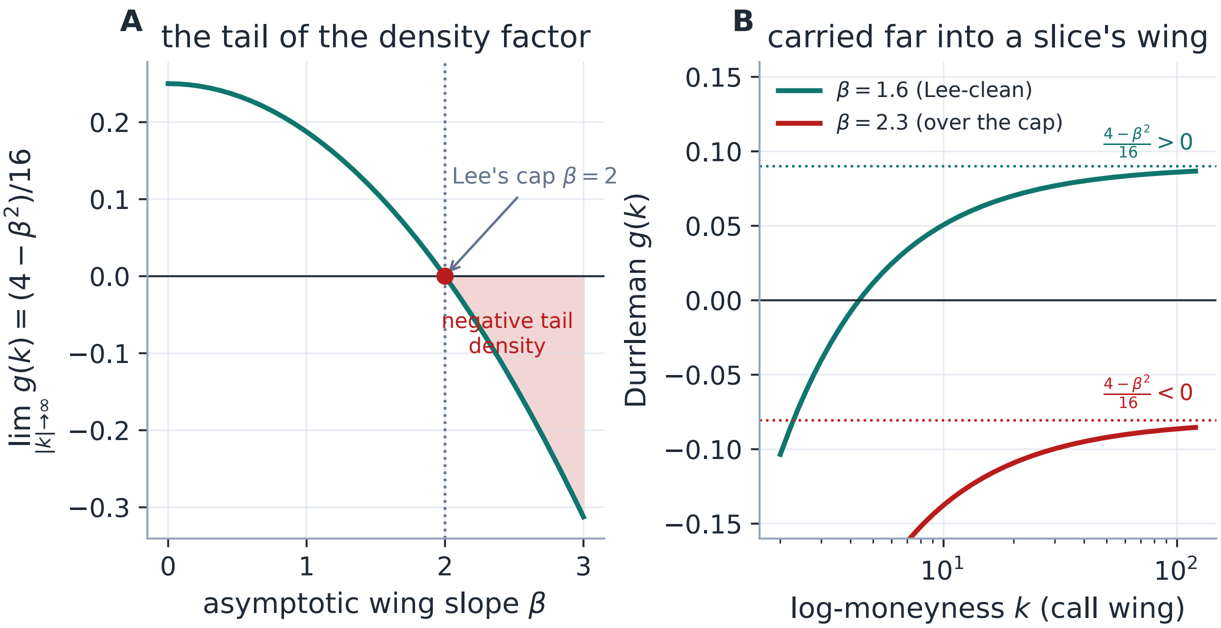

Figure 2: Lee's bound, inside the density. Panel A: the asymptotic value of the Durrleman factor is $(4-\beta^2)/16$, which changes sign exactly at $\beta = 2$; past the cap the far-tail density is negative. Panel B: two symmetric slices differing only in wing slope, followed far out — $\beta = 1.6$ rises to its positive tail limit, $\beta = 2.3$ falls to its negative one. The wing slope is not cosmetic; it fixes the sign of the density in the tail.

Caution — the boundary itself is a trap (committee revision R1, 2026-07-24)

Both screens are silent *at* the cap, and proposition 2 shows the limit decides nothing there. The slice $(a, b, \rho, m, s) = (0.04,\ 2,\ 0,\ 0,\ 0.2)$ makes it concrete: minimum total variance 0.44 (floor clean), $\beta_L = \beta_R = 2$ exactly, so a cap at 2 charges zero penalty — yet $\alpha = a - 2m = 0.04 < 2$, and by (9) $g(10) \approx -0.0485$: genuinely negative tail density behind two passing screens. This is why the production default cap is *strictly buffered*, $\beta_{\max} = 1.95 < 2$: the buffer restores "$\beta \le \beta_{\max} \Rightarrow g$ eventually positive" and excludes only laws whose moment budget beyond the first is already negligible ($p^\star = (2-\beta)^2/(8\beta) < 2 \times 10^{-4}$ at the buffered cap). Nor is the trap hypothetical: on the reference live surfaces (2026-07-18) one SPY expiry fitted under the old cap sat exactly at $b(1+|\rho|) = 2.0000$; the buffered refit lands at 1.9500 and moves no quote by more than 0.03 vol bp, while the other eleven nodes are byte-identical. The counterexample is certification-locked (`svi_lee_boundary`).

Proposition 3 (The coded screens, in JW units)**.** *Under the conversion of section 5, the two production screens read directly on the wing handles: the floor condition $w_\star \ge 0$ is $\tilde{v} \ge 0$, and Lee's cap $b(1+|\rho|) \le \beta_{\max}$ is*

$$\max(p, c)\,\sqrt{v\tau} \;\le\; \beta_{\max}\ (= 1.95).$$

Proof. The first is the definition of $\tilde{v}$ (section 5) times $\tau > 0$. For the second, $\beta_L = p\sqrt{w_0}$ and $\beta_R = c\sqrt{w_0}$ with $w_0 = v\tau$ (section 5), so $b(1+|\rho|) = \max(\beta_L, \beta_R) = \max(p, c)\sqrt{v\tau}$. ☐

So a desk can sanity-check a smile before touching the belly: floor non-negative, larger wing handle under the cap. Both are wing statements. The belly is a different matter.

Exercise 1. Show from (6) that $\beta_L + \beta_R = 2b$ and $\beta_R - \beta_L = 2b\rho$, so the two wing slopes carry exactly the same information as (b, ρ). Deduce that Lee's cap constrains only the steeper wing, and that a put-skewed slice ($\rho < 0$) can violate it on the put side while the call side is far from the bound.

4 The belly escapes

The word "convex" hides two different second derivatives. Raw SVI has $w'' > 0$ by construction (proposition 1). A butterfly spread, however, tests the convexity of the option *price* in strike, which is the sign of g in (7), and g mixes w, w' and w''. In the wings these agree — g's sign is Lee's — but in the belly they need not. It is worth fixing four increasingly strong meanings of "valid" once, because the note turns on the gaps between them.

Tier	Mathematical statement	Production status
Structural	$b > 0$, $\lvert\rho\rvert < 1$, $s > 0$	hard, by reparametrization (section 6)
Tail-clean	$w_\star \geq 0$ and $\max(\beta_L, \beta_R) < 2$ (strict)	the two soft coded screens, fenced at $\beta_{\max} = 1.95$; zero on a clean slice
Butterfly-clean	$g(k) \geq 0$ for all k (with the call boundary)	measured, not certified by the two screens
Calendar-clean	$w_{T_1}(k) \leq w_{T_2}(k)$ for $T_1 < T_2$	soft model-agnostic hinge (Note 10); not structural

Table 2: Four tiers of "valid." The first is hard, the second soft and confined to the tails, and the last two require conditions no single raw coefficient box supplies. The counterexample below sits in the gap between tiers two and three: tail-clean, butterfly-dirty.

That the tail screens do not reach tier three is not a worry in the abstract — it is a specific, classical slice. Axel Vogt's example, reproduced as Example 3.1 by Gatheral and Jacquier [2], is

$$(a, b, \rho, m, s) = (-0.0410,\ 0.1331,\ 0.3060,\ 0.3586,\ 0.4153).$$

Its minimum total variance is 0.0116 (floor clean), its larger Lee coefficient is 0.174 (far under the cap of 1.95), and by (8) *both* tails of g sit at $\approx 0.25 > 0$: the slice is impeccable in the wings. Yet g dips to -0.033 near $k \approx 0.88$ — in the belly, between the two clean tails — so the Black density is negative there and the slice is genuinely arbitrageable (fig. 3). Strict convexity of w did not protect strike-convexity of the price.

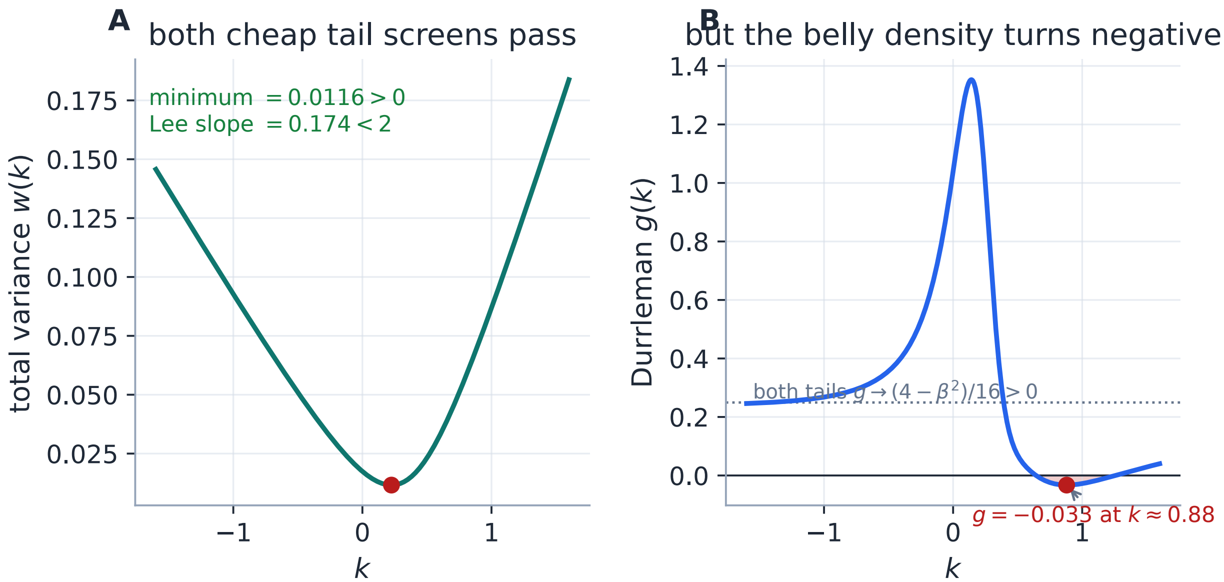

Figure 3: The honesty centrepiece. Panel A: Axel Vogt's slice is positive, strictly convex, with a positive minimum and a Lee slope far under the cap — both cheap tail screens pass. Panel B: its Durrleman factor approaches the positive tail limits $(4 - \beta^2)/16$ on both sides (dotted), yet turns negative in the belly. The screens fence the tails; the interior escapes them.

The exact butterfly-free domain of raw SVI is known — Martini and Mingone [5] characterize it completely — but it is not one short inequality: the test involves parameter rescaling, root finding and a numerical minimization. Vol-Fitter's calibrator does not implement that certified domain inside the optimizer. It instead *bounds, measures, certifies, and repairs or rejects* in a stack of five layers, at increasing distance from the core fit:

1. **Core screens** (always in the objective): the floor and Lee rows of eq. (10), finite-weight, zero on a tail-clean slice.

2. **Extrapolated-region enforcement** (opt-in, default off; Notes 09–10): tapered rows penalizing sampled $g < 0$ on the slice's own envelope, a tapered calendar hinge against the previous displayed slice, and a wing-order hinge. Only this small block is finite-differenced, so the fit runs a hybrid Jacobian.

3. **Advisory diagnostics** (always on): the Quality view measures and reports the remaining extrapolated-region $g < 0$ per fit — measurement, not enforcement.

4. **Publish-time wing projection** (export only; Note 09): a discrete projection of the exported wing samples that raises prices only, leaving the fitted core pinned.

5. **Belly certificate + acceptance rule** (committee revision R2, always on): a dense post-fit Durrleman-g certificate over the *traded* range — the region layers 1–4 cannot reach — computed from the model's own derivatives in $\sim$0.05 ms. An uncertified belly fails readiness and *hard-blocks* publish (HTTP 409); the published family additionally carries a calendar audit proving the wing projection introduced no crossings. An uncertified slice cannot become a mark.

The optimizer itself still roams the unconstrained chart — a trial iterate may be arbitrageable — but a slice can no longer LEAVE the system as a mark without passing the certificate: layers 1–4 bound the tails, measure the belly, and clean the published boundary; layer 5 is the gate.

How g is measured is part of the guarantee. An earlier backtest metric evaluated g by finite-differencing *reconstructed option prices*, and its numerical noise flagged 28.3% of provably-clean LQD slices as arbitrageable. Recomputed analytically from each model's own (w, w', w''), the LQD rate reads 0.0%, and SVI's flagged rate fell from 20.8% to 9.2% — the survivors being genuine belly violations of the kind in fig. 3. The production rule that came out of that audit: the diagnostic is always computed on model derivatives, never by differencing prices (Note 09 gives the cross-model treatment). It is a reminder that a badly *measured* g can manufacture arbitrage that isn't there, just as a tail-only screen can miss arbitrage that is.

4.1 The two soft screens, precisely

The optimizer adds to the data residual two rows that are exactly zero on a tail-clean slice and grow linearly past the boundary:

$$r_{\text{core}}(\theta) = W \begin{pmatrix} \max\big(-(a + b\,s\,q),\ 0\big) \\ \max\big(b(1+|\rho|) - \beta_{\max},\ 0\big) \end{pmatrix}, \qquad q = \sqrt{1-\rho^2}, \tag{10}$$

with residual multiplier $W = 1000$ and cap $\beta_{\max} = 1.95$. The first forbids negative total variance at the belly's floor; the second is Lee's wing bound. Because both vanish on a clean fit, a tail-clean smile is byte-identical with or without them: they are a feasibility fence, not a regularizer.

> **Caution**
>
> Read W for what it is. It multiplies the *residual* rows, so the least-squares objective receives an effective W^2 on the squared violation; and the two rows do not even share a unit — one is a total variance, the other a dimensionless slope — so W is a residual multiplier, not a "weight in vol^2." It is deliberately large enough to dominate a violated constraint and invisible on a clean one. It is not a fit-quality dial: raising it cannot improve a feasible fit, and lowering it only lets the optimizer wander into the arbitrageable cone. Both fences are also adjustable rather than unconditional — the weight can be set to zero, or the cap raised to 2 and beyond, re-opening the boundary trap of proposition 2 — so they are configuration, not guarantees.

5 Trader coordinates that split tail from belly

Raw (a, b, ρ, m, s) are excellent computational coordinates and poor market language: no single one is "the level" or "the skew," and, as fig. 4 shows, three of them move a tail and the belly at once. A desk wants to name the smile by its features: where ATM volatility sits, how steeply it leaves the money, how heavy each wing is, and where it bottoms. The jump-wing parametrization does exactly this, and — the reading this note presses — it sorts the five numbers into *two tail handles and three belly handles*.

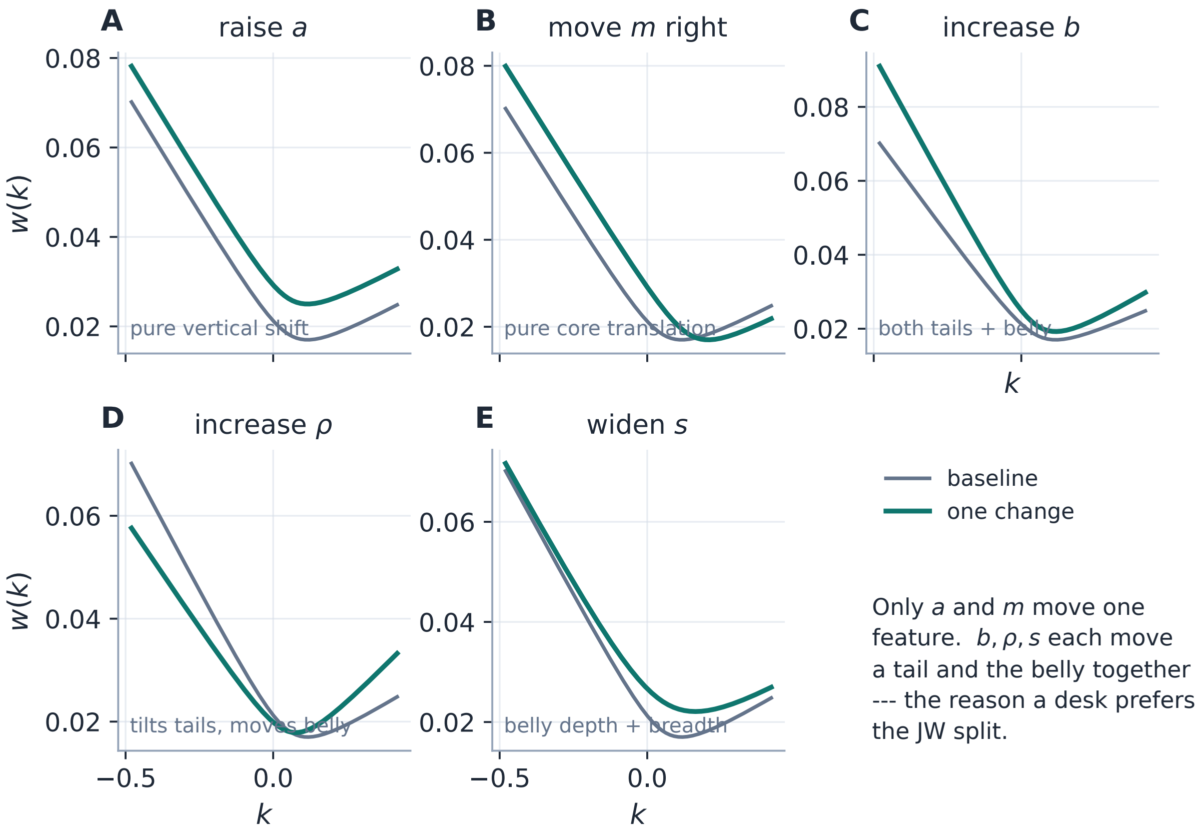

Figure 4: Why raw coordinates are poor market language. Only a (vertical shift) and m (core translation) move a single feature. Each of b, ρ, s moves a wing *and* the belly together, so a trader cannot dial one market observable without disturbing others — the motivation for a second chart.

Definition 1 (SVI-JW handles)**.** At variance clock τ, with $w_0 = w(0)$ and $w_\star = \min_k w$, the jump-wing coordinates of a slice are

$$\underbrace{v = \frac{w_0}{\tau}, \quad \psi = \big(\sqrt{w}\big)'(0), \quad \widetilde{v} = \frac{w_\star}{\tau},}_{\text{belly handles}} \qquad \underbrace{p = \frac{\beta_L}{\sqrt{w_0}}, \quad c = \frac{\beta_R}{\sqrt{w_0}}}_{\text{tail handles}}. \tag{11}$$

Substituting (2) gives the classical closed forms [2, Sec. 3.3]:

$$v = \frac{w_0}{\tau}, \quad \psi = \frac{b}{2\sqrt{w_0}}\Big(\rho - \frac{m}{\sqrt{m^2+s^2}}\Big), \quad p = \frac{b(1-\rho)}{\sqrt{w_0}}, \quad c = \frac{b(1+\rho)}{\sqrt{w_0}}, \quad \widetilde{v} = \frac{a + b\,s\,q}{\tau}. \tag{12}$$

The split is not cosmetic. The three belly handles are read on the implied-volatility chart, and the two tail handles are read on the total-variance chart, in different units:

$$I(0) = \sqrt{v}, \quad I'(0) = \frac{\psi}{\sqrt{\tau}}, \quad I(k_\star) = \sqrt{\widetilde{v}}, \qquad \beta_L = p\sqrt{v\tau}, \quad \beta_R = c\sqrt{v\tau}. \tag{13}$$

Thus $v, \widetilde{v}$ are *variances* (the IV chart shows their square roots); ψ is the ATM slope of *total volatility* $\sqrt{w}$, equal to $\sqrt{\tau}$ times the working-IV slope, *not* the plotted tangent; and p, c are normalized *total-variance* wing slopes, not slopes of the IV curve. Figure 5 annotates a fitted slice in exactly these units, with the tail handles on the total-variance panel where they live.

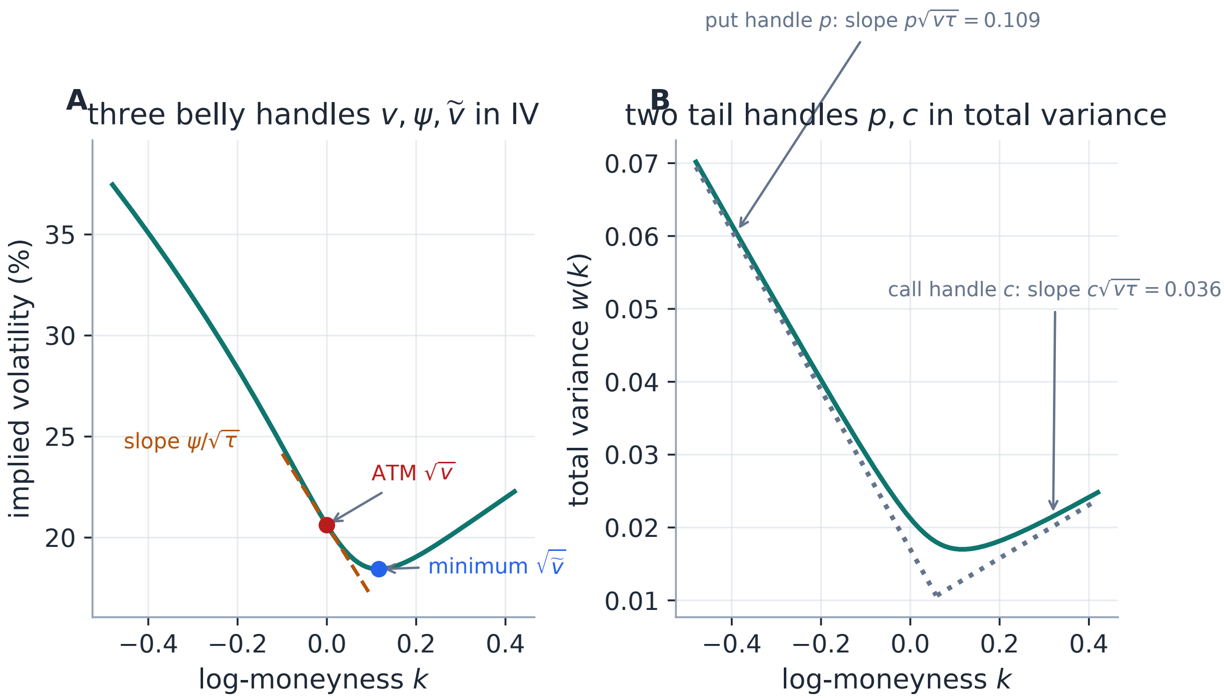

Figure 5: The five handles in their honest units. Panel A (IV): the three belly handles — the level $\sqrt{v}$, the minimum $\sqrt{\tilde{v}}$, and the ATM tangent of slope $\psi/\sqrt{\tau}$. Panel B (total variance): the two tail handles, the asymptote slopes $p\sqrt{v\tau}$ and $c\sqrt{v\tau}$. Plotting all five on one IV axis would give the wing handles units they do not have.

Handle	Reads	Functional	Visible as	Common mistake
v	belly	$w(0)/\tau$	ATM IV is $\sqrt{v}$	treating v as a volatility
ψ	belly	$(\sqrt{w})'(0)$	IV tangent is $\psi/\sqrt{\tau}$	calling ψ the plotted slope
$\tilde{v}$	belly	$w_\star/\tau$	minimum IV is $\sqrt{\tilde{v}}$	assuming positivity from the name
p	tail	$\beta_L/\sqrt{w_0}$	put slope $p\sqrt{v\tau}$	reading it on the IV axis
c	tail	$\beta_R/\sqrt{w_0}$	call slope $c\sqrt{v\tau}$	reading it on the IV axis

Table 3: Three belly handles and two tail handles, in three related but distinct languages: annualized variance, total-volatility skew, and normalized total-variance wings. Keeping the units explicit prevents nearly every JW misreading.

5.1 Inverting the handles, and where the belly information is lost

Reading JW from raw is evaluation. The reverse — reconstructing a whole hyperbola from five measurements — is the interesting direction, and it is the one the shipped converter `jw_to_raw` computes (today its only repository caller is a benchmark test; see the product-status note below). The two tail handles fix the scale and tilt,

$$b = \frac{\sqrt{w_0}}{2}(p+c), \qquad \rho = \frac{c-p}{c+p}, \qquad w_0 = v\tau, \tag{14}$$

the ATM slope fixes the normalized displacement

$$\chi := \frac{m}{\sqrt{m^2+s^2}} = \rho - \frac{4\psi}{p+c}, \tag{15}$$

and the belly gap $v - \widetilde{v}$ then sets the width. Writing

$$D(\rho,\chi) = \frac{1-\rho\chi}{\sqrt{1-\chi^2}} - \sqrt{1-\rho^2}, \tag{16}$$

the remaining coordinates are

$$s = \frac{(v-\widetilde{v})\,\tau}{b\,D(\rho,\chi)}, \qquad m = \frac{\chi\, s}{\sqrt{1-\chi^2}}, \qquad a = \widetilde{v}\tau - b\,s\sqrt{1-\rho^2}, \tag{17}$$

because $\sqrt{m^2+s^2} = s/\sqrt{1-\chi^2}$, so subtracting $w_\star = \widetilde{v}\tau$ from $w_0 = v\tau$ leaves $b\,s\,D$. The denominator D has a clean geometric form: setting $\rho = \sin\alpha$, $\chi = \sin\gamma$,

$$D = \frac{1-\cos(\alpha-\gamma)}{\cos\gamma} \;\geq\; 0, \tag{18}$$

with equality iff $\chi = \rho$. This single observation gives both the domain and the singularity.

Theorem 1 (The full smooth image of the JW map)**.** *Fix $\tau > 0$ and restrict raw SVI to $b > 0$, $|\rho| < 1$, $s > 0$, $w_0 > 0$.*

1. *A JW point has a unique regular inverse iff*

$$v > 0, \quad p > 0,\ c > 0, \quad -\frac{p}{2} < \psi < \frac{c}{2}, \quad \psi \neq 0, \quad \widetilde{v} < v, \tag{19}$$

   and then (14)–(17) return it.

2. *The image also contains the singular stratum $v > 0,\ p, c > 0,\ \psi = 0,\ \widetilde{v} = v$, every point of which is represented by* infinitely many *raw slices. There are no other smooth, nondegenerate image points.*

Proof. If $p, c > 0$ then (14) gives $b > 0$ and $\rho \in (-1, 1)$; conversely a valid slice has $p, c > 0$. From (15), using $(\rho-1)(p+c) = -2p$ and $(\rho+1)(p+c) = 2c$,

$$|\chi| < 1 \iff \rho - 1 < \tfrac{4\psi}{p+c} < \rho + 1 \iff -\tfrac{p}{2} < \psi < \tfrac{c}{2},$$

which keeps $m = \chi s/\sqrt{1-\chi^2}$ finite. By (18), $D > 0$ exactly when $\chi \neq \rho$, i.e. $\psi \neq 0$; then $s > 0$ in (17) requires the numerator $v - \widetilde{v} > 0$. Each failure breaks a requirement: $p + c \leq 0$ or one of

$p, c \leq 0$ forces $b \leq 0$ or $|\rho| \geq 1$; ψ outside the band sends $|\chi| \geq 1$; $\psi = 0$ zeroes D; $\widetilde{v} \geq v$ makes $s \leq 0$. For the singular stratum, $\psi = 0$ gives $\chi = \rho$, i.e. $k_\star = 0$ by (5), so ATM *is* the minimum and $\widetilde{v} = v$. Conversely, fix v, p, c on the stratum; (14) sets b, ρ, and for *any* $s > 0$, $m = \rho s/\sqrt{1-\rho^2}$ with $a = v\tau - b\, s\sqrt{1-\rho^2}$ yields $k_\star = 0$, $w_\star = w_0 = v\tau$, and reproduces the same five handles. □

Remark 1 (The blind spot is the belly)**.** The free width s on the singular stratum is not an abstract degeneracy: it is exactly the belly curvature. On that stratum ATM sits at the minimum, and the handles pin the level v and both tails p, c but say nothing about how sharply the curve turns through its bottom. Figure 6B shows three raw bodies that agree on $(v, 0, p, c, v)$ and differ only in how rounded the belly is. The jump-wing chart is a fine coordinate system *away* from this stratum and ill-conditioned near it: by (18) the denominator vanishes *quadratically*,

$$D \sim \frac{(\chi-\rho)^2}{2(1-\rho^2)^{3/2}} = \frac{8\psi^2}{(p+c)^2(1-\rho^2)^{3/2}}, \qquad \psi \to 0, \tag{20}$$

so recovering a finite width as $\psi \to 0$ forces $v - \widetilde{v} = O(\psi^2)$, and a naive evaluation of (16) subtracts two nearly equal numbers. The reference implementation of appendix D uses an algebraically identical but cancellation-free denominator; no rearrangement can restore a curvature the handles never carried.

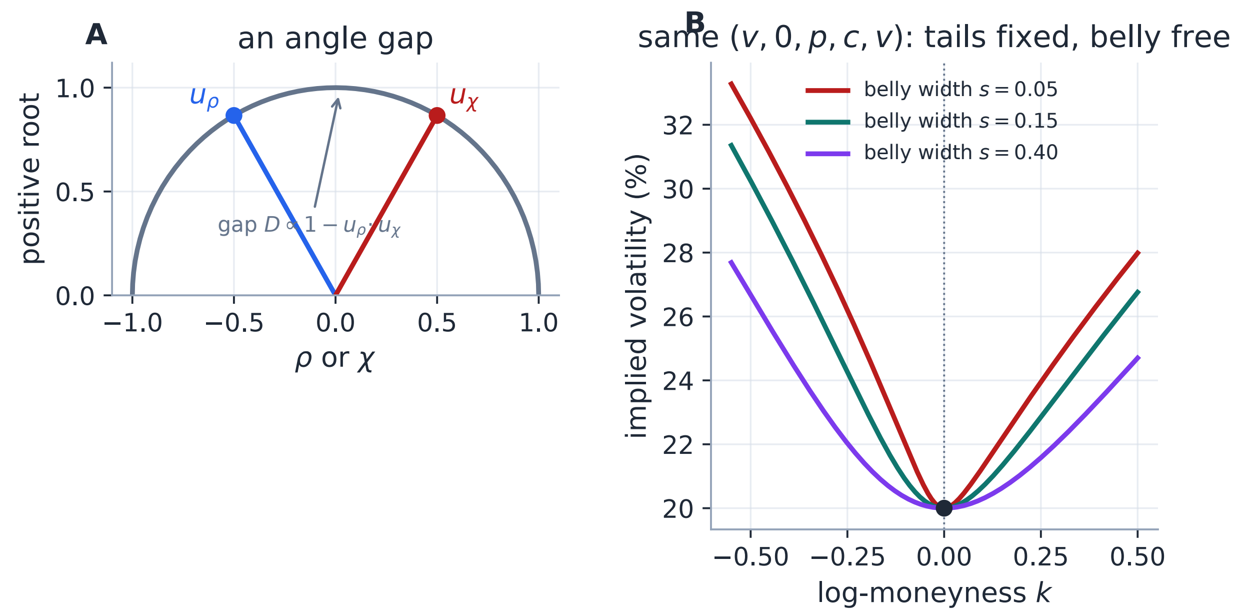

Figure 6: The $\psi = 0$ stratum is a belly blind spot. Panel A: the inverse denominator D is an angular gap between the unit vectors u_ρ and u_χ; it closes exactly when $\psi = 0$. Panel B: three raw slices with identical $(v, 0, p, c, v)$ — same level, same asymptotic rays — differing only in belly width s. The handles fix the tails and lose the interior curvature.

Caution — product status, and the unguarded converter

Production calibration fits and stores *raw* SVI as the selected overlay family and displays three model-agnostic numeric handles — ATM vol, skew and curvature by finite differences of the displayed curve — for every model. There is no backend `raw_to_jw`, no five-handle JW API/UI entry, no JW bumping and no JW export; those are candidate features, not shipped ones. The shipped `jw_to_raw` implements (14)–(17) with *zero* domain checks, and its failure modes differ by case rather than being uniformly silent: a scalar $p + c = 0$ raises a division error; ψ outside $(-p/2, c/2)$ takes the square root of a negative and returns NaNs; $\widetilde{v} \geq v$ returns $s \leq 0$, not a slice; a near-zero ψ amplifies rounding. This is deliberate — the only caller feeds it benchmark handles well inside theorem 1 — but it makes the regular domain a *contract*, now test-locked. Any future user-facing JW workflow must validate (19) first, treat the singular stratum separately, and use the stable denominator.

Exercise 2. On the singular stratum, differentiate the five functionals (12) with respect to s at fixed $(v, \psi, p, c, \widetilde{v})$ using $m = \rho s/\sqrt{1-\rho^2}$, $a = v\tau - b\,s\sqrt{1-\rho^2}$, and verify that $\partial(v, \psi, p, c, \widetilde{v})/\partial s = 0$. Conclude that the Jacobian of the raw→JW map drops rank there — an analytic statement of the belly blind spot.

6 Fitting the slice: fit the belly, fence the tails

The optimizer never handles a bounded raw vector. It receives $\theta \in \mathbb{R}^5$ and maps

$$a = \theta_1, \quad b = \operatorname{softplus}(\theta_2), \quad \rho = \tanh(\theta_3), \quad m = \theta_4, \quad s = e^{\theta_5}, \tag{21}$$

so that every finite θ is a structurally valid hyperbola (tier one of table 2). This removes the box constraints entirely: the unconstrained Levenberg–Marquardt (LM) solver never wastes a step on the boundary and never proposes a nonsensical hyperbola. It guarantees no more than structure — a stays free, so a trial slice can still carry negative total variance somewhere, which the residual handles by flooring w at 10^{-12} inside the square root (a solver device; `RawSVI.implied_vol` itself has no floor). Tail feasibility is the business of the soft rows (10); belly feasibility is not enforced at all, by design.

Objective, weights, start. The data term is a plain implied-volatility residual $r_i = \sqrt{\omega_i}\,(I(k_i) - I_i)$; bid–ask and haircut modes replace it by a vol-space band hinge with a small mid anchor. The surfaced quote-weight choices are `equal` and `tv_density` (Note 07), the same two every model gets; the low-level API also accepts arbitrary ω_i (e.g. vega^2, used by research scripts), which is a code capability, not a setting. The initializer is geometric and reads the belly and the wings directly: the argmin of the quoted w seeds (a, m), and two finite-span wing slopes seed (b, ρ) through (6), after which (21) is inverted for θ_0. On a liquid smile a single LM pass converges.

6.1 The analytic Jacobian, in three layers

The residual Jacobian is worth deriving, because it is what makes SVI the speed peer of the other models. It is three chain-rule layers. First, the raw-parameter partials of the hyperbola, with x, r from (3):

$$\frac{\partial w}{\partial a} = 1, \quad \frac{\partial w}{\partial b} = \rho x + r, \quad \frac{\partial w}{\partial \rho} = b\,x, \quad \frac{\partial w}{\partial m} = -b\Big(\rho + \frac{x}{r}\Big), \quad \frac{\partial w}{\partial s} = \frac{b\,s}{r}. \tag{22}$$

Second, the data rows live in IV, so the Black chain rule for $I = \sqrt{w/\tau}$:

$$\frac{\partial I}{\partial w} = \frac{1}{2\sqrt{w\tau}} = \frac{1}{2\tau I}, \tag{23}$$

zeroed wherever the $w \leq 10^{-12}$ floor is active. Third, the reparametrization (21):

$$\frac{\mathrm{d}b}{\mathrm{d}\theta_2} = \operatorname{sigmoid}(\theta_2) = 1 - e^{-b}, \qquad \frac{\mathrm{d}\rho}{\mathrm{d}\theta_3} = 1 - \rho^2, \qquad \frac{\mathrm{d}s}{\mathrm{d}\theta_5} = s. \tag{24}$$

The hinge rows follow the subgradient convention "active linear part, else zero": the floor row differentiates $-(a + b\,s\,q)$, the Lee row differentiates $b(1 + |\rho|)$ with the $\operatorname{sgn}(\rho)$ subgradient (zero at $\rho = 0$), and each is a zero row where its violation is not strictly positive. The calibrator uses this closed form whenever no var-swap or prior block is present; those blocks are not differentiated and revert the fit to finite differences, exactly as in LQD. Under extrapolation enforcement the fit is *hybrid*: the rows above stay analytic and only the small extrapolation block is finite-differenced.

Performance

The solver is Levenberg–Marquardt, not trust-region reflective: on noisy real chains LM crosses the penalty kinks in far fewer iterations at matched tolerance (`trf` at 10^{-10} was measured slower on real nodes), so the analytic Jacobian is a same-convergence drop-in that swaps scipy's $1 + P$ finite-difference evaluations ($P = 5$) for one closed-form call. Two measurements, labelled for what they are. (i) This note's private residual/Jacobian microbenchmark on the 25-quote synthetic case (section 7), warmed median of five, single-threaded on the development machine, core rows only: 3.12 ms → 1.30 ms, a 2.40× speed-up with the two objective costs agreeing to 2.7×10^{-33} (same solution within fit precision, not bit-identical). (ii) A historical real-node result: spike-regime backtest nodes, June-2026 harness, same optimizer and tolerances, 26.3 ms → 10.2 ms per fit (≈ 2.58×) — the finding that motivated shipping it. See appendix B.

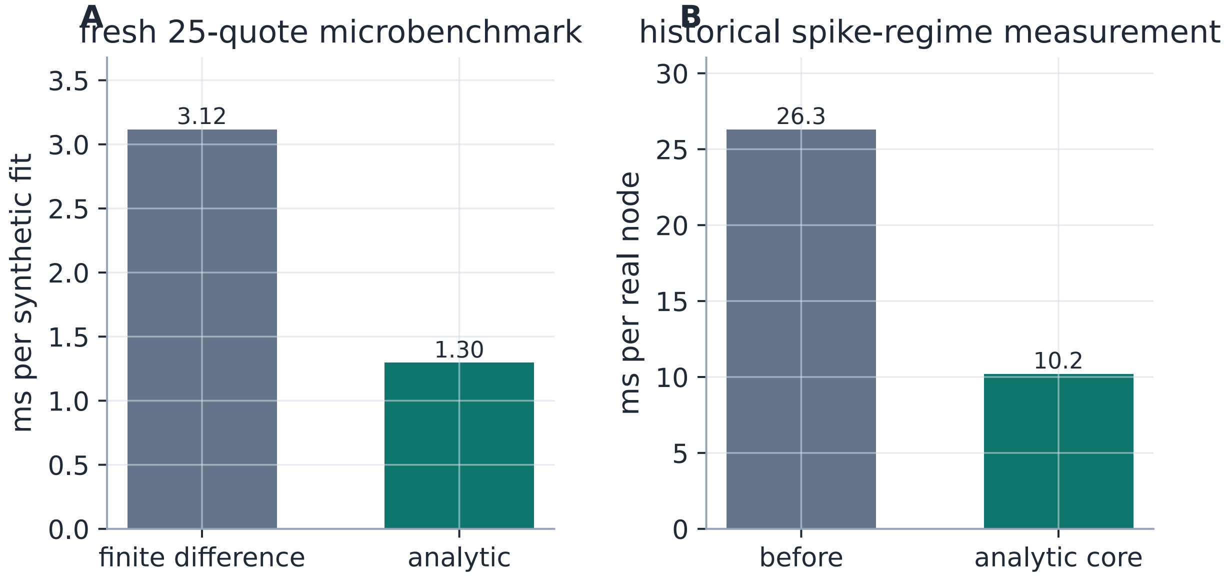

Figure 7: The saving is fewer residual evaluations, not a different optimizer. Panel A is the fresh synthetic microbenchmark; Panel B the historical real-node measurement. The two have different scales and provenance and isolate the same implementation change; machine-dependent numbers and protocol live in appendix B.

6.2 The optional blocks

The optional blocks share the series' product targets through model-specific residuals, and each is byte-identical to its own absence:

- **Band fit** (Note 07): the mid residual becomes a vol-space band hinge with a small mid anchor.
- **Calendar hinge** (Note 10): the model-agnostic $\sqrt{W_{\text{cal}}}\,\max(\text{floor} - w(k), 0)$ against the nearer expiry — the same target LQD expresses through its quantile, written here on w. A finite-weight hinge drives crossings down, not provably to zero.
- **Extrapolation fences** (default off; Notes 09–10): the tapered rows of section 4, hybrid-Jacobian when on.
- **Var-swap target** (Note 08) and **prior persistence** (Note 13): the same targets as LQD through SVI-native residuals, so the SVI overlay now receives the prior treatment the primary model does — closing an old asymmetry in which SVI overlays got no prior at all.

7 Worked example: a laboratory round trip

An exact-family, noise-free recovery is not market evidence; it is the right laboratory test for a coordinate conversion and a Jacobian, since any visible miss is an implementation error. Take $\tau = 0.5$ and the jump-wing handles $(v, \psi, p, c, \widetilde{v}) = (0.0425, -0.25, 0.75, 0.25, 0.034)$. The *production* `jw_to_raw` builds the target raw slice $(a, b, \rho, m, s) \approx (0.010625, 0.07289, -0.5, 0.05831, 0.10100)$; we sample 25 noise-free quotes on $[-0.35, 0.30]$, refit production raw SVI from the geometric start, and read the handles back — a genuine JW→raw→fit→JW loop whose forward leg is shipped code.

Report two errors, and keep them separate. The *quote* error — the report-space miss — is a maximum of 5.6×10^{-13} vol bp over the 25 quotes, in 21 residual evaluations. The *latent* error — the miss in the object we actually recovered — is a maximum jump-wing round-trip deviation of 3.9×10^{-16} across all five handles (table 5 reproduces the target to display precision). The recovered wing slopes are $\beta_L = 0.1093$ (put) and $\beta_R = 0.0364$ (call). Both errors are at machine precision because the target *is* an SVI slice: the exercise validates the optimizer and the exact conversion, and says nothing about SVI's expressiveness on a non-SVI market. On the diagnostic grid g stays above 0.343 (fig. 8C).

Table 4: Recovered raw coordinates.

Parameter	Value
a	+0.010625
b	+0.072887
ρ	-0.500000
m	+0.058310
s	+0.100995

Table 5: Recovered SVI-JW handles.

Handle	Value
v	+0.042500
ψ	-0.250000
p	+0.750000
c	+0.250000
$\widetilde{v}$	+0.034000

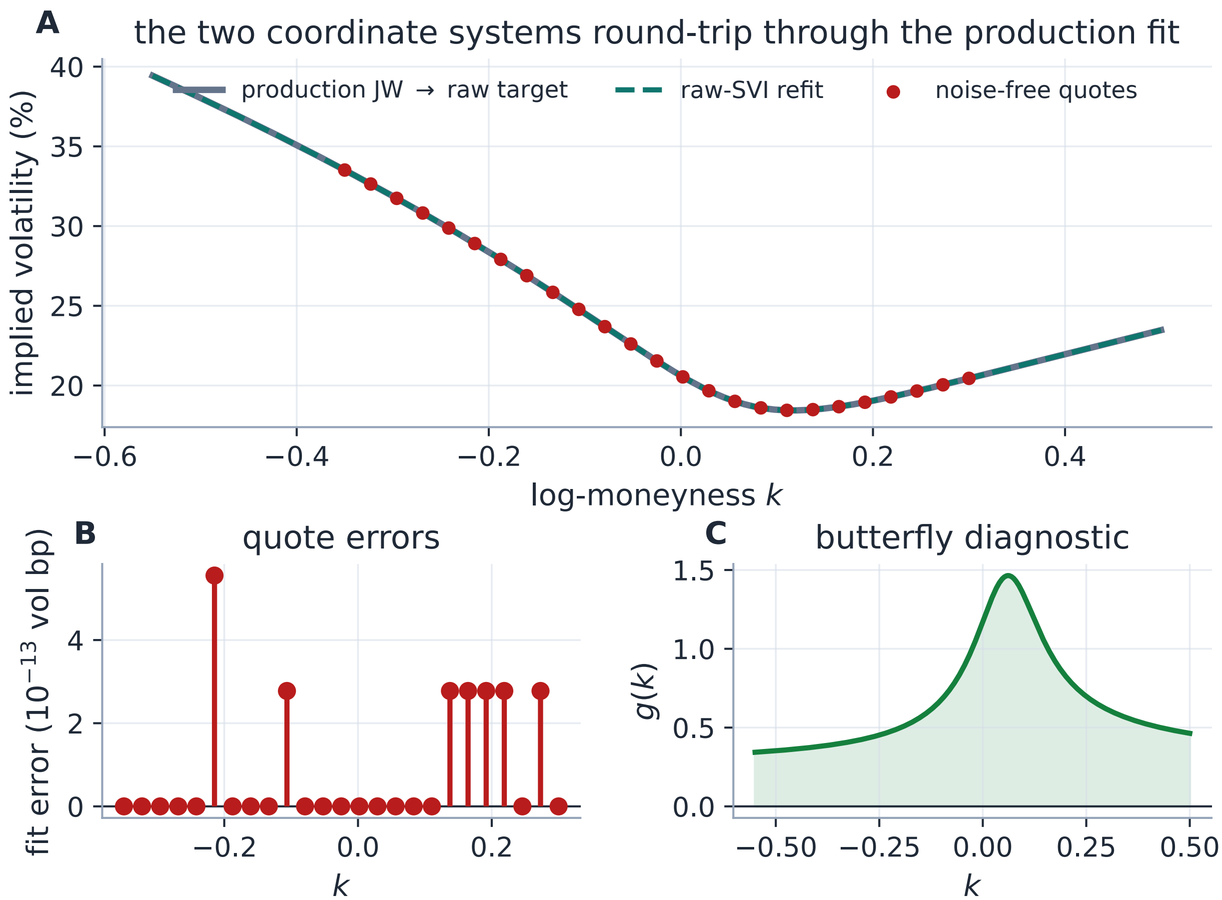

Figure 8: The two coordinate systems round-trip through the production fit. Panel A: target and refit coincide, so the useful evidence is the residual scale (Panel B, in 10^{-13} vol bp) and the independent Durrleman diagnostic (Panel C). The test exercises conversion, initialization, calibration and reverse readout in one reproducible path; it deliberately says nothing about real-market fit.

> **Heuristic**
>
> On real NBBO history the published spike-regime replay measured SVI at 24.3 vol bp RMS in-sample and 26.8 out-of-sample across $\approx 1,576$ nodes per regime — an order of magnitude worse than the laboratory number above, and the honest figure of merit. SVI is also stiffer than LQD or Local-Vol on event or double-hump shapes, for the reason section 8 makes precise. (Those replay parquets are historical and not checked into this workspace; the numbers are retained for continuity with the note and deck.)

8 One belly, one turn

Strict convexity, $w'' > 0$, buys SVI one minimum and one monotone turn of the slope from $-\beta_L$ to $+\beta_R$. On a liquid equity-index chain this is exactly the right regularization: it forbids the wiggles that noise would otherwise carve into a smile. But it is also a hard expressiveness ceiling. A W-shaped total-variance target (Note 03), a double-humped event density (Note 01's double-hat), or a sharply localized short-dated kink asks the curve to turn more than once, and one hyperbola has exactly one belly. Figure 9 makes the failure quantitative: against a smooth, positive, deliberately two-minimum target, production raw SVI returns the best single-belly compromise, missing by 147.6 vol bp RMS and 351.9 vol bp at worst.

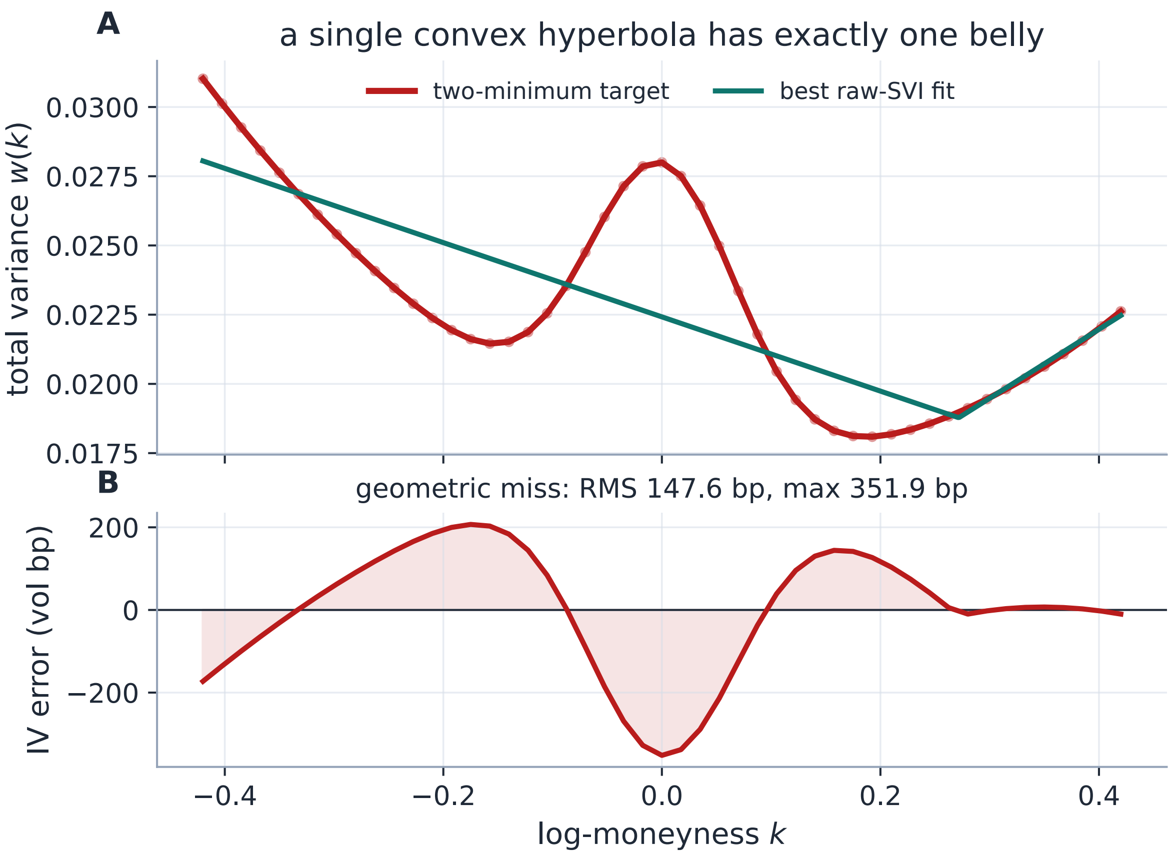

Figure 9: A geometric stress test, not a market smile. The synthetic target has two minima and a central hump; one convex hyperbola cannot follow it and settles for the best compromise (A), leaving the structured residual of Panel B. This is the price of the convexity that regularizes ordinary smiles.

Sparse identification is a second, subtler limitation, and it is where the tail/belly reading pays off once more. On a narrow or one-sided quote window the initializer's "wing slopes" are finite-span proxies read off the quoted range, not asymptotic observations, and distinct raw tuples can price the quoted strikes almost identically while implying different remote wings. The fitted *curve* can be stable even when (a, b, ρ, m, s) is not. JW is easier to read, but its tail handles p, c are still *extrapolated* functionals when the market has not quoted the wings. Across fits, compare prices, curves, and observed handles — not raw coefficients.

Exercise 3. The two-minimum target of fig. 9 is positive and smooth but not convex. Argue directly from $w'' > 0$ that no raw SVI slice can have two local minima, and hence that the RMS miss cannot be driven to zero by any choice of (a, b, ρ, m, s) — the limitation is structural, not a matter of optimizer tuning.

9 What is genuinely original, and limitations

SVI, the jump-wing handles, and the arbitrage analysis are classical: raw SVI and JW are due to Gatheral, the arbitrage-free theory to Gatheral–Jacquier and Durrleman, the wing bound to Lee, and the exact butterfly domain to Martini–Mingone. The reading offered here — *wings as a moment reading of the tails, belly as the interior the cheap screens cannot reach* — is a framing, not a theorem. The implementation choices worth flagging are three: *structural validity by reparametrization* (21), so the solver cannot leave tier one except through the two soft tail fences; the *exact raw↔JW conversion with a proved regular domain and an identified belly blind spot* (theorem 1, remark 1); and the *analytic LM Jacobian* that makes SVI the speed peer of the other overlays.

Limitations, stated plainly, because a lecture that sells a model without drawing its boundary is an advertisement. Raw SVI total variance is one globally convex hyperbola with a single belly, so it cannot reproduce a W-shaped smile and can over-smooth a sharp short-dated skew (section 8). It is butterfly-clean only over a cone the coded fences do not certify (section 4), and carries no calendar coupling on its own — that is supplied, softly, by the model-agnostic hinge of Note 10. On sparse or one-sided chains its raw parameters can be poorly identified even where the curve is stable, so comparisons should be made on curves and handles, never on (a, b, ρ, m, s). And the jump-wing chart, elegant as it is, tears on the $\psi = 0$ stratum, where it keeps the tails and forgets the belly.

A Hyperparameter atlas

Table 6: SVI and shared calibration controls.

Knob	Default	Role
Surfaced (FitSettings / OptionsSettings)		
`model`	`lqd`	Set to `svi` to display the raw-SVI overlay.
`sviPenaltyWeight` W	1000	Residual multiplier of the two tail rows (10) (effective W^2 in the squared cost; 0 = fences off).
`leeSlopeMax` $\beta_{\max}$	1.95	Lee wing-slope cap on $b(1+\lvert\rho\rvert)$; buffered strictly below 2 by default (the boundary itself admits negative tail density); values ≥ 2 are accepted but re-open the trap.
`weightScheme`	`equal`	Per-quote weights, `equal` / `tv_density` (Note 07); arbitrary weights exist only at the low-level API.
`midAnchorWeight`	0.05	Mid anchor in the band-fit objective (Note 07).
`haircut`	0.005	Absolute-vol tightening, haircut band mode only.
`calendarWeight`	10^6	Calendar hinge weight (Note 10; squared objective, residual uses its root).
`extrapEnforce`	off	Tapered extrapolated-region fences (section 4); calibration-affecting, hybrid Jacobian when on.
Internal (reparametrization / solver)		
$b = \text{softplus}(\theta_2)$	—	Enforces $b > 0$.
$\rho = \tanh(\theta_3)$	—	Enforces $\lvert\rho\rvert < 1$.
$s = e^{\theta_5}$	—	Enforces $s > 0$.

Knob	Default	Role
LM tolerances	10^{-15}	`xtol/ftol/gtol`; LM converges in few iterations, so tight tolerances are cheap.
trial-w floor	10^{-12}	Inside $\sqrt{w/\tau}$, to keep the IV residual evaluable on structurally-valid-but-negative-variance trials; not part of `RawSVI.implied_vol`.

B Performance notes

1. **Levenberg–Marquardt over trust region.** On real noisy chains LM crosses the penalty kinks in fewer iterations than `trf` at matched tolerance; `trf` at 10^{-10} was measured slower on real nodes.

2. **Analytic Jacobian.** Closed-form Jacobian of the core residual stack (data + tail penalties + calendar) through the reparametrization, with the two scopes stated: the note's microbenchmark (25 synthetic quotes, core rows only, single-threaded development machine, warmed median of five) measured 1.30 ms vs 3.12 ms (2.40×) with objective costs agreeing to 2.7×10^{-33} — same solution within fit precision, not bit-identical; the historical real-node measurement (spike-regime backtest nodes, June-2026 harness) was 26.3 ms → 10.2 ms (≈ 2.58×). Machine-dependent numbers belong here, not in the body. Gated to the var-swap/prior-free configuration; hybrid (analytic core + FD extrapolation rows) under `extrapEnforce`.

3. **Structural validity by construction.** The reparametrization removes box-constraint handling, so each LM step is a plain linear solve.

4. **Geometric warm start.** Reading (a, m) from the belly and (b, ρ) from the wing slopes lands θ_0 close enough that liquid smiles converge in a single pass.

5. **Converter conditioning.** The reference inverse of appendix D uses the cancellation-free denominator of remark 1; it still rejects the singular stratum, because no rearrangement can select a missing belly curvature.

C Traceability

A word on what these anchors prove, so they are not over-read. The benchmark-fit tests exercise an already tail-clean target, so the "enforcement" rows lock that clean fits are untouched and that the penalty rows and their Jacobians are correct when active — they do not demonstrate that an adversarial fit ends feasible. The recovery test whose name says "machine precision" asserts parameter agreement at $\sim 10^{-4}$ relative and curve agreement at $\sim 10^{-6}$; the freshly generated example reaches machine precision, but the lock is looser than its name. A finite diagnostic grid is not a global proof.

Table 7: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor *Test anchors*
Wing slopes are $b(1 \mp \rho)$; tail limit of g is $(4-\beta^2)/16$	proposition 1, proposition 2	`models/svi_jw/svi.py::RawSVI.wing_slopes` `backtest/dispatch.py::_analytic_butterfly`
JW→raw conversion exact on the benchmark	section 5	`models/svi_jw/svi.py::jw_to_raw` `test_lqd_pricing.py::test_jw_conversion_matches_note`

Regular inverse domain; $\psi = 0$ non-identification; invalid-input behaviour	theorem 1, remark 1	`models/svi_jw/svi.py`
		`test_svi_domain.py::test_regular_domain_round_trips`
		`::test_psi_zero_stratum_not_identified`
		`::test_invalid_inputs_fail_loudly_not_plausibly`

Tail screens do not certify butterfly freedom (Axel Vogt slice)	section 4	`models/svi_jw/calibrate.py::_penalties` `test_svi_domain.py::test_core_screens_do_not_certify_butterfly_freedom`

Noise-free benchmark recovery (params $\sim 10^{-4}$ rel, curve $\sim 10^{-6}$)	section 7	`models/svi_jw/calibrate.py`
		`test_svi_calibrate.py::test_recovers_benchmark_parameters`
		`::test_curve_reproduced_to_machine_precision` (name predates the looser lock)

Clean fits untouched by the fences (benchmark is tail-clean)	proposition 3	`models/svi_jw/calibrate.py::_penalties`
		`test_svi_calibrate.py::test_respects_lee_wing_bound`
		`::test_min_variance_non_negative`

Analytic Jacobian matches FD, penalty rows active or not	section 6.1	`models/svi_jw/jacobian.py`
		`test_svi_jacobian.py::test_mid_fit_admissible`
		`::test_lee_penalty_active`
		`::test_min_variance_penalty_active`

Calendar hinge rows correct; off is byte-identical	section 6	`models/svi_jw/calibrate.py`
		`test_svi_jacobian.py::test_calendar_floor_active`
		`test_overlay_calendar.py::test_svi_no_floor_is_byte_identical`

Prior blocks reach the SVI overlay	section 6	`models/svi_jw/calibrate.py` `test_prior_parametric.py::test_operator_prior_pulls_all_models_toward_prior_skew`

D Reference implementation: the two maps

The raw slice evaluates in one line, $w(k) = a+b(\rho(k-m)+\sqrt{(k-m)^2+s^2})$; the maps below read the handles off it and invert them. The listing is imported and executed by `figures/gen_svi_moments.py`, which compares the checked inverse against production `jw_to_raw` on three regular-domain points to relative tolerance 2×10^{-12} before any figure is written — so the code printed here is the code that produced the note's numbers. The inverse uses the cancellation-free denominator of remark 1; the reverse map is exactly the functional definition (12).

Listing 1: Executable raw/JW reference maps used by this note (`figures/svi_moments_reference.py`).

```python
"""Executable reference maps for the moment-bound edition of Note 02.

These two functions are the note's Appendix D listing.  ``raw_to_jw`` reads the
five jump-wing functionals off a raw slice (splitting the smile into its two
tail handles ``p, c`` and its three belly handles ``v, psi, vtilde``);
``jw_to_raw_checked`` is a domain-guarded, cancellation-resistant inverse.  The
generator executes both against the production converter before drawing a single
figure, so the listing printed in the note is the code that produced its numbers.
"""

from __future__ import annotations

import numpy as np

from volfit.models.svi_jw.svi import RawSVI, SVIJW


def raw_to_jw(raw: RawSVI, tau: float) -> dict[str, float]:
    """Read the five JW functionals off a raw slice.

    ``p, c`` are the two *tail* handles (normalized asymptotic wing slopes);
    ``v, psi, vtilde`` are the three *belly* handles (ATM level, ATM slope of
    total volatility, and the floor).
    """
    w0 = float(raw.total_variance(0.0))
    root0 = np.sqrt(raw.m * raw.m + raw.sigma * raw.sigma)
    sqw0 = np.sqrt(w0)
    return {
        "v": w0 / tau,
        "psi": raw.b * (raw.rho - raw.m / root0) / (2.0 * sqw0),
        "p": raw.b * (1.0 - raw.rho) / sqw0,
        "c": raw.b * (1.0 + raw.rho) / sqw0,
        "v_tilde": (raw.a + raw.b * raw.sigma * np.sqrt(1.0 - raw.rho**2)) / tau,
    }


```

```python
def jw_to_raw_checked(jw: SVIJW) -> RawSVI:
    """Domain-guarded regular inverse with a cancellation-resistant denominator.

    The two tail handles fix the scale ``b`` and tilt ``rho``; the ATM slope
    fixes the normalized displacement ``chi``; the belly gap ``v - vtilde`` then
    sets the width ``sigma``.  The denominator ``D`` vanishes quadratically as
    ``psi -> 0`` (the belly blind spot), so it is evaluated in a form free of the
    catastrophic cancellation of the textbook expression.
    """
    if not (
        jw.t > 0.0 and jw.v > 0.0 and jw.p > 0.0 and jw.c > 0.0
        and -0.5 * jw.p < jw.psi < 0.5 * jw.c
        and jw.psi != 0.0 and jw.v_tilde < jw.v
    ):
        raise ValueError("JW point is outside the regular inverse domain")
    w0 = jw.v * jw.t
    b = 0.5 * np.sqrt(w0) * (jw.p + jw.c)
    rho = (jw.c - jw.p) / (jw.c + jw.p)
    chi = rho - 4.0 * jw.psi / (jw.p + jw.c)
    q_rho, q_chi = np.sqrt(1.0 - rho * rho), np.sqrt(1.0 - chi * chi)
    # D = (1 - rho*chi)/q_chi - q_rho, rearranged so no two large near-equal
    # numbers are subtracted; algebraically identical, numerically stable.
    dq = (chi - rho) * (chi + rho) / (q_rho + q_chi)
    denom = ((rho - chi) ** 2 + dq**2) / (2.0 * q_chi)
    width = (w0 - jw.v_tilde * jw.t) / (b * denom)
    m = chi * width / q_chi
    a = jw.v_tilde * jw.t - b * width * q_rho
    return RawSVI(a=float(a), b=float(b), rho=float(rho),
                  m=float(m), sigma=float(width))
```

References

[1] J. Gatheral. A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives. Presentation, Global Derivatives, Madrid, 2004.

[2] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.

[3] R. W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004.

[4] V. Durrleman. From implied to spot volatilities. *Finance and Stochastics*, 14(2):157–177, 2010.

[5] C. Martini and A. Mingone. No arbitrage SVI. *SIAM Journal on Financial Mathematics*, 13(1):227–261, 2022.